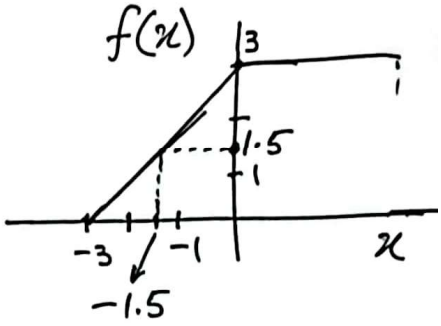
SOLUTION – MID TERM EXAM

CALCULUS AND ANALYTICAL GEOMETRY

FALL – 2024

Q1 (a) From the graph,

$$f(-1.5) = 1.5$$



(b) $$f(x) = \begin{cases} x+3 & -3 \le x < 0 \\ 3 & x \ge 0 \end{cases}$$

(c) Domain : $[-3, \infty)$ OR $[-3, 3]$

Range : $[0, 3]$

(d) From the graph, $f(0) = 3$

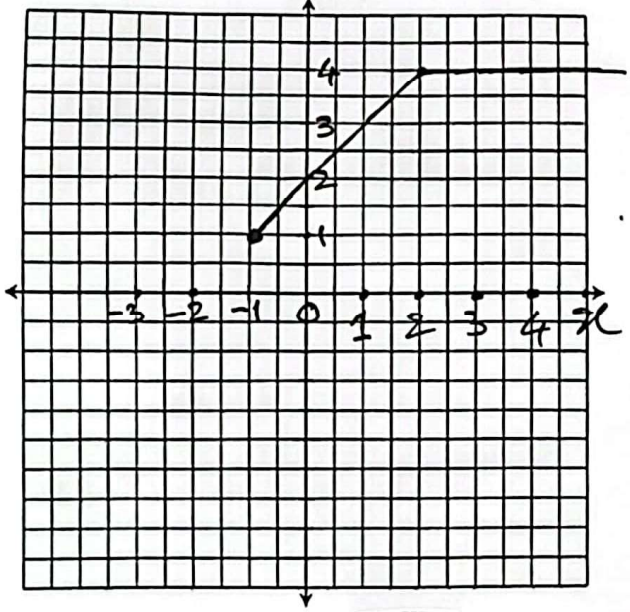
Thus, $g(f(0)) = g(3) = 3^2 = 9$

(e) Solution on graph paper.

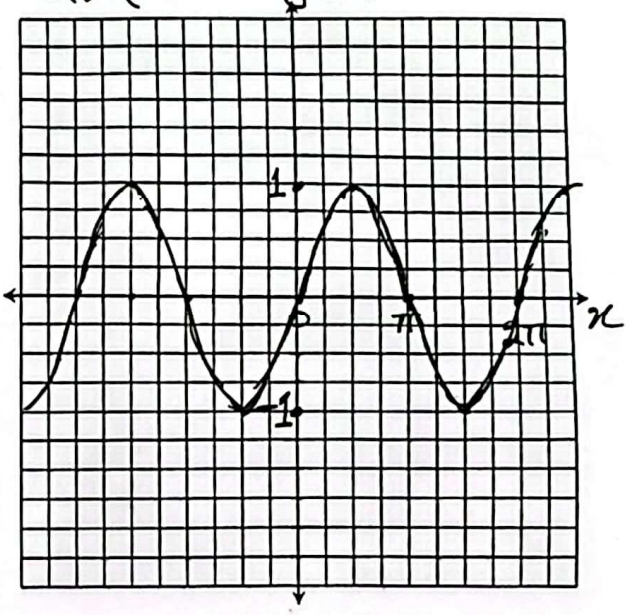
Q2 Solution on graph paper.

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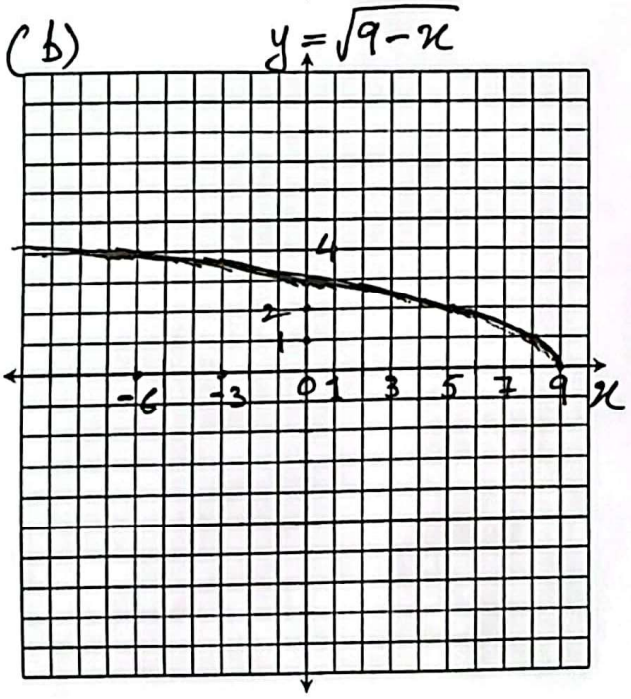
Q1(e)

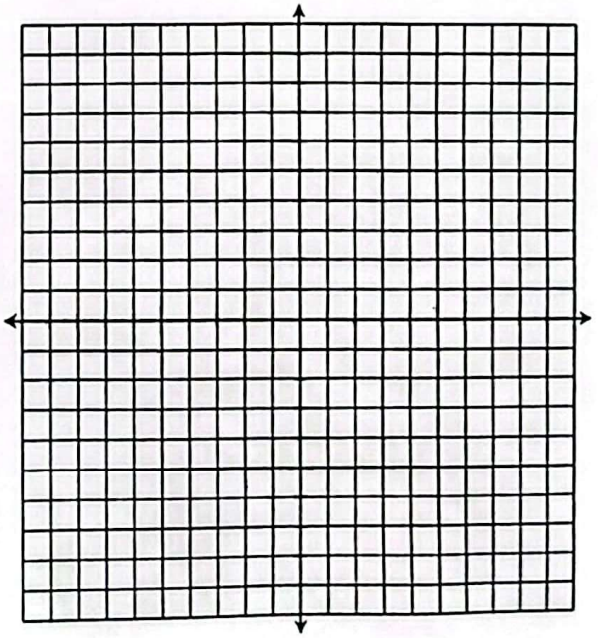


Q2(a)



Q2(b)





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Q3 (a) $\lim_{x\to -5} \frac{x^2+3x-10}{x+5} = \lim_{x\to -5} \frac{(x+5)(x-2)}{(x+5)}$

$= \lim_{x\to -5} (x-2) = -5-2 = -7$

(b) $\lim_{v\to 2} \frac{v^3-8}{v^4-16} = \lim_{v\to 2} \frac{(v-2)(v^2+2v+4)}{(v-2)(v+2)(v^2+4)}$

$= \lim_{v\to 2} \frac{v^2+2v+4}{(v+2)(v^2+4)} = \frac{2^2+2(2)+4}{(2+2)(2^2+4)} = \frac{4+4+4}{4\times 8}$

$= \frac{12}{32} = \frac{3}{8}$

(c) $\lim_{x\to -1} \frac{\sqrt{x^2+8}-3}{x+1} = \lim_{x\to -1} \frac{\sqrt{x^2+8}-3}{x+1} \times \frac{\sqrt{x^2+8}+3}{\sqrt{x^2+8}+3}$

$= \lim_{x\to -1} \frac{(\sqrt{x^2+8})^2 - 3^2}{(x+1)(\sqrt{x^2+8}+3)} = \lim_{x\to -1} \frac{x^2+8-9}{(x+1)(\sqrt{x^2+8}+3)}$

$= \lim_{x\to -1} \frac{x^2-1}{(x+1)(\sqrt{x^2+8}+3)} = \lim_{x\to -1} \frac{(x-1)(x+1)}{(x+1)(\sqrt{x^2+8}+3)}$

$= \frac{-1-1}{\sqrt{(-1)^2+8}+3} = \frac{-2}{\sqrt{9}+3} = \frac{-2}{3+3} = -\frac{1}{3}$

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